Harshit Singh

AI Engineer | Python • PyTorch/TensorFlow • Multimodal AI • MLOps

+91-8826483132 | harshit.workmain@gmail.com | linkedin.com/in/harshit-singh-3b8467300 | github.com/harshitworkmain | portfolio-website-harshit-08.vercel.app

Education

Vellore Institute of Technology, Chennai — Chennai, India
*B.Tech, Electronics and Computer Engineering — CGPA: **7.97*** — *2023 – 2027*

New Era School (CBSE) — Ghaziabad, Uttar Pradesh
*Class XII: **87.16%** — Class X: **89.83%*** — *2020 – 2023*

Experience

AI & Robotics Intern — Autonomous Mobile Robotics [**Certificate**] [**GitHub**] — Jul 2025 – Aug 2025
STEMTEC AI & Robotics Technology Pvt. Ltd. — *Chennai, India*

- Engineered a **15 Hz** stochastic state-space estimation pipeline (Extended Kalman Filter, ROS 2 `robot_localization`) fusing wheel-encoder odometry with 6-DOF IMU telemetry, cutting mean spatial drift by **60%** (**0.45 m** → **0.18 m**) and orientation error by **69.2%** (**7.8°** → **2.4°**) across 10 m ground-truth trials.
- Built a real-time closed-loop reactive policy processing ∼**3,600 points/sec** of LiDAR telemetry at **10 Hz** via 60° frontal-sector range slicing, achieving a **95% collision-free navigation success rate** (19/20 trials) at **sub-50 ms** end-to-end inference latency.

Projects

SPARC — Edge ML & Multi-Modal Perception Engine *(PyTorch, YOLOv8, TensorFlow.js)* [**Demo**] [**Web-App**]

- Engineered a low-latency edge ML pipeline (<**8ms/sample**, **30 FPS**) on ARM Cortex-A72, cutting CPU compute **35%** via 229-dim landmark vectorization (**99.97%** compression).
- Orchestrated multi-model inference (YOLOv8 Nano + CNN) with a 45-frame sliding-window buffer, cutting false-positives **94%** — **Top 5 finalist** (500+ teams), SIH 2025. [**Certificate**]

TRIDENT — Predictive AI & Risk Modeling *(PyTorch, NetworkX, Flask)* [**Demo**] [**Live**] [**GitHub**]

- Architected a 2-layer stacked PyTorch LSTM (`hidden_size=50`, dropout **0.2**, Adam η=0.001) forecasting 24-hour microclimate trends at **sub-0.005 MSE** across **17,500+** hourly records via sliding-window sequences.
- Built graph-based risk solvers (NetworkX weighted diffusion, Betweenness/Closeness centrality, Edmonds-Karp max-flow, Bellman-Ford) and deployed via Flask/Gunicorn/Docker on Render, cutting container size ∼**80%** with CPU-only inference wheels.

NeuroBridge AI — Multimodal Clinical ML Platform *(MediaPipe, TensorFlow.js, Node.js)* [**Live**] [**GitHub**]

- Architected an edge CV pipeline (MediaPipe FaceLandmarker, WebGL) extracting **468** 3D landmarks at **30 FPS** in-browser, feeding a multimodal late-fusion clinical risk model.
- Deployed and served a TensorFlow.js NLP toxicity classifier (τ=**0.85**, 7 categories) in production with resilient model serving (keep-alive + backoff), sustaining **99.9%** uptime.

DrishtiGuide — Edge CV & Assistive Perception System *(PyTorch, OpenCV, YOLOv8)* [**GitHub**]

- Deployed a real-time YOLOv8n pipeline (buffer flushing, 3-frame subsampling), cutting compute overhead **66.7%** at **sub-100ms** end-to-end latency.
- Built an automated LBPH retraining pipeline (zero manual cold-start) and a 6-DOF IMU fall-detection classifier with monocular distance estimation (≤**3cm** error).

Skills

Languages: Python, C, SQL, Java
ML/DL Frameworks: PyTorch, TensorFlow/Keras, OpenCV, Scikit-Learn, CUDA
Multimodal AI, CV & NLP: YOLOv8, MediaPipe, LSTM/Time-Series Forecasting, NLP Classification, Multimodal Late Fusion, Edge Inference Optimization
MLOps & Systems: AWS, Docker, CI/CD, Flask/Gunicorn, ROS 2, Linux (Ubuntu), Bash, Git, REST APIs
Professional Skills: Data-Driven Problem Solving, Cross-Functional Collaboration, Communication, Time Management

Achievements & Leadership

- **3rd Place, Technovation 2025** (Kurukshetra '25, CEG Guindy, Anna University) — ranked **2nd Runner Up out of 50+** teams; **INR 15,000** award. [**Certificate**]
- **Winner, Innovate X Impact (V-NEST)** — secured **1st Position out of 35** teams; **INR 5,000** award.
- **Deputy Operations Lead, Team Namo Nirvana (VIT Chennai)** — led **100+ member** teams across **5–6 competitions/semester**, boosting efficiency by **35%** and securing **2 sponsors.** [**LOA**] [**Certificate**] [**LOR**]

Languages

English (Fluent) | **Hindi** (Native)